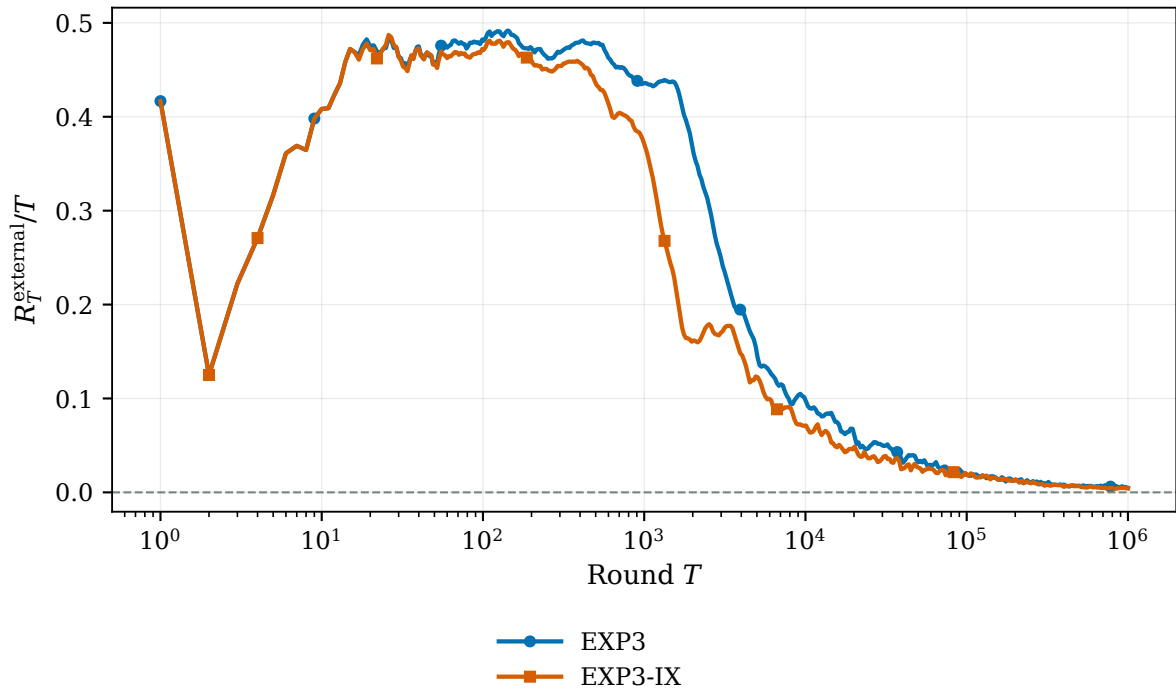


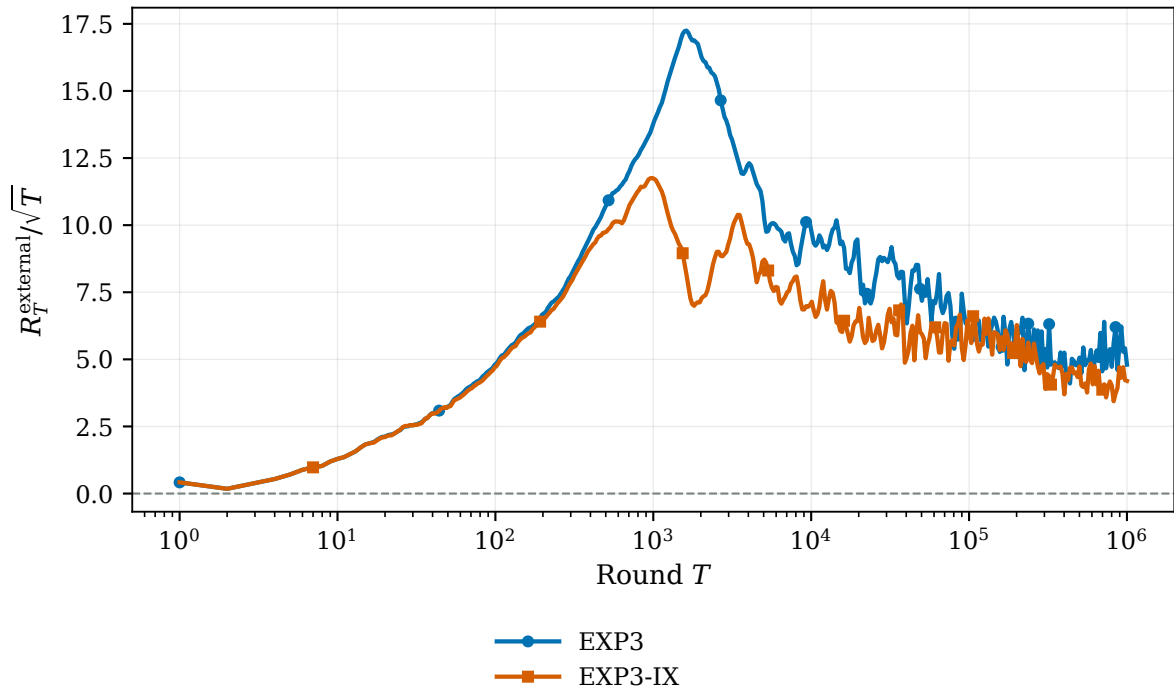
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 0.004803, SD = 0.00170114, SE = 0.000491078
EXP3-IX:	mean = 0.00419725, SD = 0.0025064, SE = 0.000723535



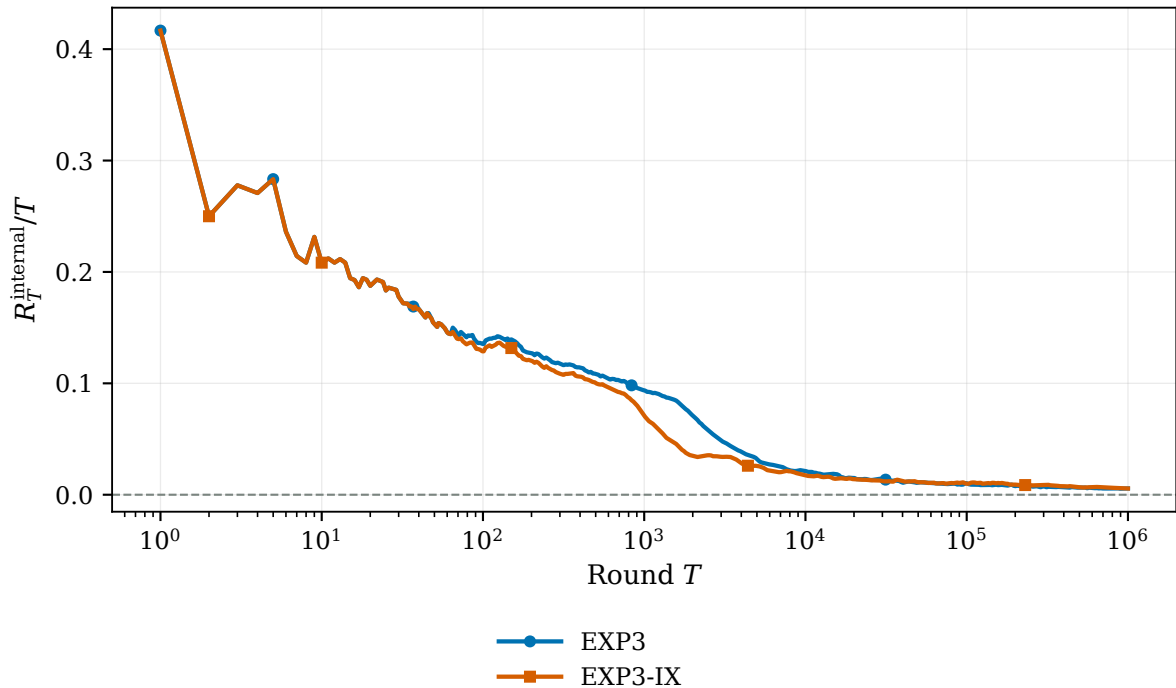
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 4.803, SD = 1.70114, SE = 0.491078
EXP3-IX:	mean = 4.19725, SD = 2.5064, SE = 0.723535



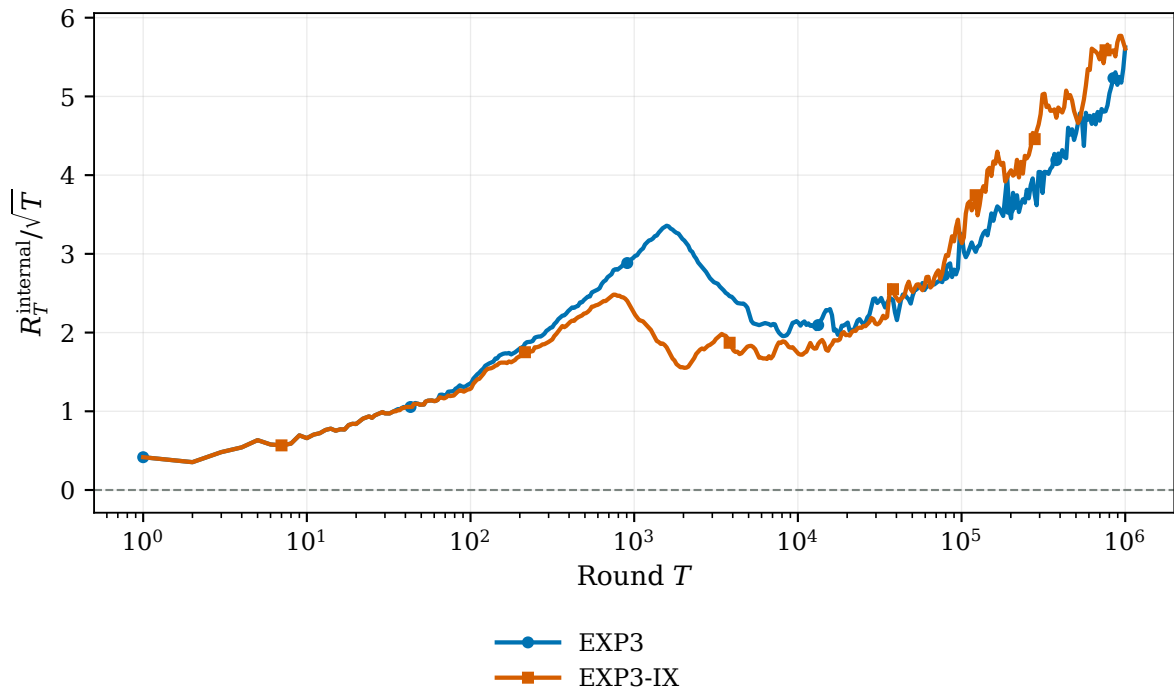
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 0.00562192, SD = 0.00148798, SE = 0.000429542
EXP3-IX:	mean = 0.00560775, SD = 0.00108382, SE = 0.000312871



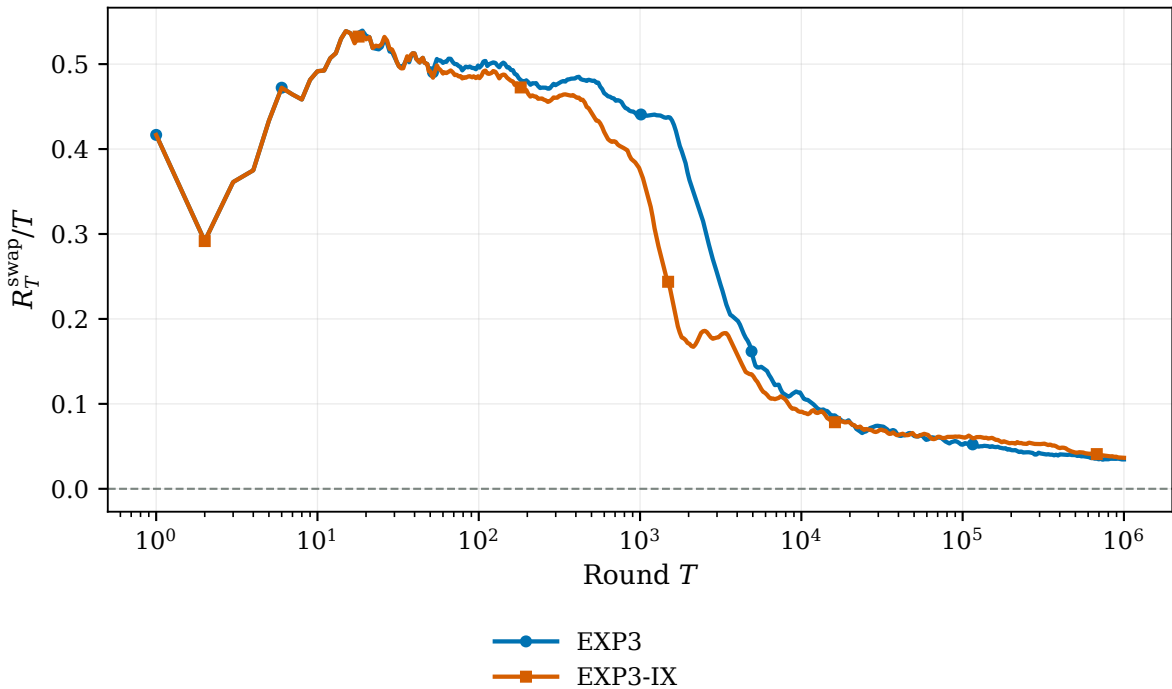
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 5.62192, SD = 1.48798, SE = 0.429542
EXP3-IX:	mean = 5.60775, SD = 1.08382, SE = 0.312871



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 0.0346296, SD = 0.00799226, SE = 0.00230717
EXP3-IX:	mean = 0.036513, SD = 0.00761423, SE = 0.00219804



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Bandit feedback
Algorithms:	EXP3, EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3:	mean = 34.6296, SD = 7.99226, SE = 2.30717
EXP3-IX:	mean = 36.513, SD = 7.61423, SE = 2.19804

